



The Carnac stones in Brittany, France, are a series of more than 3,000 upright megaliths. The oldest stones date from around 4500 BCE.



Spinning threads

Spindle whorls are often the first evidence of spinning—spun threads are wrapped around a spindle shaft. Whorls are usually light; if heavier than 5 oz (150g), they tend to break the thread.

South Asia by 5500 BCE; throughout Europe by 3000 BCE; and as far as China and Southeast Asia by 2000 BCE.

Casting metal objects with a mold developed in the 5th millennium BCE. The first known cast metal object comes from Mesopotamia and dates from about 3200 BCE.

Spinning raw fibers to make a thread may have begun as early as the 7th millennium BCE, which is the date of spindle whorls found at Çatal Höyük, Turkey. **Weaving** may have arisen from the late Paleolithic skill of making nets and baskets. The loom—a frame or brace to keep one set of threads (the warp) tense while another (the weft) is interwoven with it—appeared in the form of warp beams (simple sticks) and backstrap looms (the warp beams were held taut by a strap around the user's back) in West Asia and Egypt by the 4th millennium BCE.

During the early years of agriculture, ground for sowing had to be cultivated using **handheld digging sticks** or hoes. The use of cattle as draft animals made the eventual use of **the ard, or scratch plow**, possible. This primitive wooden plow, sometimes with a metal tip, cut shallow furrows. The earliest evidence of its use comes from the 4th millennium BCE, and it spread widely in Egypt, West Asia, and Europe.

The quality of ceramics was improved by the invention, in around 6000 BCE, of **kilns**—specially built chambers in which pottery could be fired. **Two-chamber, updraft kilns** (in which the fire is in the lower chamber) appeared in the Hassuna culture of Mesopotamia in about 6000 BCE. Around

“ **BARLEY IS THRESHED FOR YOU, WHEAT IS REAPED FOR YOU, YOUR MONTHLY FEASTS ARE MADE WITH IT, YOUR HALF-MONTHLY FEASTS ARE MADE WITH IT.** ”

Ancient Egyptian pyramid text, c.2400–2300 BCE

5000 BCE, the method of “coiling” pots was improved by a simple turntable (tournette) beneath the pot. By 3500 BCE, the tournette had been replaced in southern Mesopotamia by a **true potter's wheel**, consisting of a heavy stone wheel that could be turned rapidly and continuously. This allowed the potter to throw the pot by placing a lump in the center of the device and shaping it as the wheel spun around.

For many millennia, all transportation on land was by foot. The first artificial aids were sleds, which have been found in Finland dating from 6800 BCE, and skis, in use in Russia around 6300 BCE. The **invention of the wheel** revolutionized transportation.

Wagon in clay

This clay pot in the shape of a wagon dates from around 3000 BCE and shows the typical features of early wheeled vehicles from central and southern Europe.

Four-wheeled wagons appeared in Poland and the Balkans around 3500 BCE and soon afterward in Mesopotamia. At first, wheels were solid disks connected to the wagon by a wooden axle, but around 2000 BCE, **spoked wheels were developed**, which made lighter, more mobile vehicles possible. Around 3100 BCE,

efficient harnesses for attaching draft animals to wagons were developed in Mesopotamia, allowing greater loads and distances to be attained.

As commercial transactions grew more complex, accurate measurements of goods became essential. **Standardized weight and length measures** were introduced in Mesopotamia, Egypt, and the Indus Valley in the late 4th millennium BCE. The earliest weights were often based on grains of wheat or barley, which have a uniform weight. The standard unit of length, the cubit, was based on the length of a man's forearm.

